Dear Dr. Wake,

Please consider this manuscript "Phenological sensitivity to climate change disrupts species coexistence" as a Perspective in *Nature Climate Change*.

Shifts in phenology—the seasonal timing of organisms' life cycle events—are one of the most widespread biological indicators of climate change, yet the strength of these shifts vary substantially among species (1; 2). These phenological differences among taxa have often been overlooked in ecological theory, but growing evidence suggests that they may play a major role in dictating community interactions and shaping the structure and function of ecological communities (3; 4; 5). While this recognition has spurred advances towards integrating phenology into community assembly models, current frameworks are not designed to capture the dynamic interactions between the physical environment, phenology and competition (6). As a result research to date has struggled to make inference on how species interactions will shift with climate change (7).

Here, we highlight a major gap in current approaches that limits progress in both empirical phenological research and community assembly. While most work has focused on phenological differences between species as an inherent attribute, here we show how phenological differences are better defined as the realized product of an interaction between the environment (cues) and species physiologies. This product in turn structures varying competition each year, depending on the environment, and provides a new framework to understand phenology within coexistence theory. Integrating examples from empirical data on seed germination phenology, as well as established community assembly models, we outline how modeling **differences in species' phenological sensitivity** (i.e., their phenological response or reaction norm to environmental variation) in place of the expressed phenological differences themselves can overcome current limitations. With this subtle, yet essential, adjustment we review how phenologically-mediated patterns of coexistence under historical climate conditions may degrade as climate warms while providing a biologically realistic mechanism to project changes in these species iterations into the future.

Importantly, our framework highlights clear steps to move the field forward. Species-level estimates of phenological sensitivity can be readily quantified with simple experiments, and our proposed modeling approaches can be leveraged to incorporate multiple, interacting environmental cues, variable environmental stressors, and more complex species interactions. Taken together, these advances could rapidly develop predictive frameworks that can ultimately be used to inform species conservation and ecological management. Our manuscript details a path forward to do this effectively and advance ecological theory, climate change biology, and the translational ecology that links them together.

We have previous discussed this manuscript with *Nature Climate Change* Editor Dr. Tegan Armarego-Marriott during a recent meeting of Ecological Society of America meeting (Long Beach, California), and feel it would be of broad interest to the readership of *Nature Climate Change*. The main text of this manuscript is 2001 words in length and it contains two figures. It is co-authored by M.J. Donahue and E.M. Wolkovich and is not under consideration elsewhere. We hope that you will find it suitable for publication in *Nature Climate Change*, and look forward to hearing from you.

Sincerely,

Daniel Buonaiuto

References

[1] Zhang, H., Chuine, I., Regnier, P., Ciais, P., and Yuan, W. Deciphering the multiple effects of climate warming on the temporal shift of leaf unfolding. *Nature Climate Change* **12**(2), 193–199 (2022).

[2] Roslin, T., Antão, L., Hällfors, M., Meyke, E., Lo, C., Tikhonov, G., Delgado, M. d. M., Gurarie, E., Abadonova, M., Abduraimov, O., Adrianova, O., Akimova, T., Akkiev, M., Ananin, A., Andreeva, E., Andriychuk, N., Antipin, M., Arzamascev, K., Babina, S., Babushkin, M., Bakin, O., Barabancova, A., Basilskaja, I., Belova, N., Belyaeva, N., Bespalova, T., Bisikalova, E., Bobretsov, A., Bobrov, V., Bobrovskyi, V., Bochkareva, E., Bogdanov, G., Bolshakov, V., Bondarchuk, S., Bukharova, E., Butunina, A., Buyvolov, Y., Buyvolova, A., Bykov, Y., Chakhireva, E., Chashchina, O., Cherenkova, N., Chistjakov, S., Chuhontseva, S., Davydov, E. A., Demchenko, V., Diadicheva, E., Dobrolyubov, A., Dostoyevskaya, L., Drovnina, S., Drozdova, Z., Dubanaev, A., Dubrovsky, Y., Elsukov, S., Epova, L., Ermakova, O., Ermakova, O. S., Ershkova, E., Esengeldenova, A., Evstigneev, O., Fedchenko, I., Fedotova, V., Filatova, T., Gashev, S., Gavrilov, A., Gaydysh, I., Golovcov, D., Goncharova, N., Gorbunova, E., Gordeeva, T., Grishchenko, V., Gromyko, L., Hohryakov, V., Hritankov, A., Ignatenko, E., Igosheva, S., Ivanova, U., Ivanova, N., Kalinkin, Y., Kaygorodova, E., Kazansky, F., Kiseleva, D., Knorre, A., Kolpashikov, L., Korobov, E., Korolyova, H., Korotkikh, N., Kosenkov, G., Kossenko, S., Kotlugalyamova, E., Kozlovsky, E., Kozsheechkin, V., Kozurak, A., Kozyr, I., Krasnopevtseva, A., Kruglikov, S., Kuberskaya, O., Kudryavtsev, A., Kulebyakina, E., Kulsha, Y., Kupriyanova, M., Kurbanbagamaev, M., Kutenkov, A., Kutenkova, N., Kuyantseva, N., Kuznetsov, A., Larin, E., Lebedev, P., Litvinov, K., Luzhkova, N., Mahmudov, A., Makovkina, L., Mamontov, V., Mayorova, S., Megalinskaja, I., Meydus, A., Minin, A., Mitrofanov, O., Motruk, M., Myslenkov, A., Nasonova, N., Nemtseva, N., Nesterova, I., Nezdoliy, T., Niroda, T., Novikova, T., Panicheva, D., Pavlov, A., Pavlova, K., Podolski, S., Polikarpova, N., Polyanskaya, T., Pospelov, I., Pospelova, E., Prokhorov, I., Prokosheva, I., Puchnina, L., Putrashyk, I., Raiskaya, J., Rozhkov, Y., Rozhkova, O., Rudenko, M., Rybnikova, I., Rykova, S., Sahnevich, M., Samoylov, A., Sanko, V., Sapelnikova, I., Sazonov, S., Selyunina, Z., Shalaeva, K., Shashkov, M., Shcherbakov, A., Shevchyk, V., Shubin, S., Shujskaja, E., Sibgatullin, R., Sikkila, N., Sitnikova, E., Sivkov, A., Skok, N., Skorokhodova, S., Smirnova, E., Sokolova, G., Sopin, V., Spasovski, Y., Stepanov, S., Stratiy, V., Strekalovskaya, V., Sukhov, A., Suleymanova, G., Sultangareeva, L., Teleganova, V., Teplov, V., Teplova, V., Tertitsa, T., Timoshkin, V., Tirski, D., Tolmachev, A., Tomilin, A., Tselishcheva, L., Turgunov, M., Tyukh, Y., Van, P., Van, V., Vasin, A., Vasina, A., Vekliuk, A., Vetchinnikova, L., Vinogradov, V., Volodchenkov, N., Voloshina, I., Xoliqov, T., Yablonovska-Grishchenko, E., Yakovlev, V., Yakovleva, M., Yantser, O., Yarema, Y., Zahvatov, A., Zakharov, V., Zelenetskiy, N., Zheltukhin, A., Zubina, T., Kurhinen, J., and Ovaskainen, O. Phenological shifts of abiotic events, producers and consumers across a continent. *Nature Climate Change* **11**(3), 241–248 (2021).

[3] Cleland, E. E. and Wolkovich, E. Effects of phenology on plant community assembly and structure. *Annual Review of Ecology, Evolution, and Systematics* **55**(Volume 55, 2024), 471–492 (2024).

[4] Alexander, J. M. and Levine, J. M. Earlier phenology of a nonnative plant increases impacts on native competitors. *Proceedings of the National Academy of Sciences* **116**(13), 6199–6204, 2026/04/01 (2019).

[5] Rudolf, V. H. W. The role of seasonal timing and phenological shifts for species coexistence. *Ecology Letters* **(2019).**

[6] Loughnan, D., Joly, S., Legault, G., Kharouba, H. M., Betancourt, M., and Wolkovich, E. M. Phenology varies with phylogeny but not by trophic level with climate change. *Nature Ecology & Evolution* 8(10), 1889–1896 (2024).

[7] Urban, M. C., Bocedi, G., Hendry, A. P., Mihoub, J. B., Pe'er, G., Singer, A., Bridle, J. R., Crozier, L. G., De Meester, L., Godsoe, W., Gonzalez, A., Hellmann, J. J., Holt, R. D., Huth, A., Johst, K., Krug, C. B., Leadley, P. W., Palmer, S. C. F., Pantel, J. H., Schmitz, A., Zollner, P. A., and Travis, J. M. J. Improving the forecast for biodiversity under climate change. *Science* 353(6304), aad8466, 2026/04/01 (2016).